

RESEARCH ARTICLE

Advanced EEG Signal Processing and Deep Q-Learning for Accurate Student Attention Monitoring

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ABSTRACT Measuring student attention during learning remains a challenging task. Traditional methods, such as surveys and observations, are subjective and often inaccurate. While EEG can detect attentional changes, its interpretation is complicated by noise and artifacts. This study introduces an enhanced approach that leverages advanced data preprocessing techniques and a Double Deep Q-Network (DDQN) model, a specialized deep reinforcement learning (DRL) algorithm adept at handling complex, multidimensional EEG data. We employ wavelet transformations to generate robust frequency-based representations of raw multi-channel EEG signals, followed by Butterworth bandpass and notch filters to eliminate noise and artifacts. Dimensionality reduction and feature scaling are achieved employing Principal Component Analysis (PCA) alongside Independent Component Analysis (ICA), leading in cleaner and more representative EEG data. Our method significantly improves the accuracy of attention state classification, achieving a test accuracy of 98.4%. This advancement sets a new standard for utilizing EEG data to monitor attention in educational settings, showcasing the effectiveness of our preprocessing techniques in enhancing the DDQN's neural decoding capabilities.

INDEX TERMS Student attention, EEG signal processing, double deep q-network (DDQN), deep reinforcement learning (DRL), attention state classification, neural decoding.

I. INTRODUCTION

With the speed at which digital education is developing and the growing dependence on e-learning environments, it is critical to measure and maximise student engagement. Persistent focus and concentration greatly improve integrated learning and cognitive processing [1]. The use of EEG signatures to measure degrees of involvement and attention has been verified by earlier research [2], [3]. But most of the time, these systems are based on supervised learning methods, which need explicit attention state labels while training models [4]. An important research gap is the limited work on implicitly modeling attention fluctuations from EEG

data through reinforcement learning (RL). RL is particularly suited for handling complex temporal relationships and high-dimensional data like EEG [5].

Ur Rehman et al. [6] address the challenge of quantifying attention during learning using EEG and a DDQN algorithm. Traditional methods are subjective and inaccurate. The DDQN processes EEG data with wavelet transformations, extracting features to classify attention states (attentive, non-attentive, drowsy). This approach sets a new standard for adaptive educational systems. This current paper is an extension of our work in [6], building upon its foundation to further advance the field.

In this work, we provide a unique method to indirectly quantify attention fluctuations from EEG inputs using a DDQN. Our DDQN is taught to recognise EEG-attention

The associate editor coordinating the review of this manuscript and approving it for publication was Yongming Li¹.

interrelationships using a personalised incentive scheme, in contrast to conventional techniques that depend on explicit labelling. By dividing attention states into “attentive,” “non-attentive,” and “drowsy” categories, this method allows for more detailed modelling of attention states than the binary classifications found in earlier research [7]. With a 98% accuracy in multi-class classification, our model attains a noteworthy enhancement, signifying a 6% rise in comparison to earlier DQN models. The dual Q-learning approach and a special asymmetric reward function included in the DDQN architecture enable this progress.

Because neuroelectrical signals are small and chaotic and sometimes obscured by artefacts with considerably greater amplitudes, it is difficult to indirectly estimate attention from EEG data [8]. Furthermore, inter and intra-individual heterogeneity makes modelling more difficult to be accurate and broadly applicable [9]. Advanced deep learning architectures are necessary to overcome these constraints by revealing hidden correlations in high-dimensional signals linked to attention changes in a variety of real-world educational contexts [10].

Deep Reinforcement Learning has demonstrated remarkable efficacy in handling noisy, real-time data streams [11]. Harnessing these advantages, our work focuses on separating distinct attentional states from EEG data. The suggested DDQN model classifies three attention levels attentive, non-attentive, and drowsy with excellent accuracy using EEG characteristics. Through continuous EEG inputs and instantaneous feedback, our technology dynamically modifies digital learning environments to improve student engagement.

In order to assess and dynamically increase user engagement, real-time adaptive interfaces are needed, as this research discusses. Our RL-based technique, in contrast to conventional supervised learning models, adjusts in response to ongoing EEG data, guaranteeing a customised learning environment that keeps learners engaged without interfering with the learning process.

The key advancements of this research work are as follows:

- **An innovative deep reinforcement learning architecture:** Closes a crucial research gap in the use of RL for decoding neural correlates of attention by introducing a novel DDQN agent for multi-class categorization of attention levels using EEG characteristics.
- **Granular quantification of dynamic attention levels:** Compared to earlier benchmarks, our method achieves a substantially greater accuracy in classifying attention into three distinct levels: attentive, non-attentive, and drowsy.
- **Advanced EEG data preprocessing techniques:** Implements a comprehensive preprocessing pipeline, including wavelet transformations, butterworth band-pass and notch filters, PCA, and ICA, to improve the clarity and accuracy of EEG signals, which enhances the robustness and reliability of attention state classification.

This approach enabled us to achieve a 98.4% accuracy, a level of precision not achieved by other models of machine learning or AI.

II. RELATED WORK

The integration of EEG and deep learning techniques has shown significant promise in enhancing e-learning platforms by personalizing content based on real-time emotional state recognition. Below, we summarize the related research and identify gaps that our work addresses, as shown in Table 1.

The integration of EEG and deep learning techniques has shown significant promise in enhancing e-learning platforms by personalizing content based on real-time emotional state recognition. For instance, the author [12] utilized EEG and the K-nearest neighbors (KNN) algorithm to recognize emotional states with 74.3% accuracy, thereby enhancing student satisfaction in learning English listening and reading skills. However, this study primarily focused on supervised learning techniques, which require explicit labels for model training.

DRL has proved to be an effective approach for addressing the limitations of supervised learning in EEG analysis. The study [13] developed a novel DRL architecture for engagement estimation using video data. The model employs a customized policy, which allows it to estimate engagement levels (low, medium, high) and request human input in uncertain cases, progressively refining the policy and engagement classifier. Similarly, the authors in [14] proposed RLPS, a reinforcement learning system that pre-selects images in facial expression recognition (FER) tasks, outperforming state-of-the-art FER approaches on multiple datasets.

In the context of educational EEG applications, The research study [15] Developed a signal processing pipeline that integrates a set of filters, common spatial pattern techniques, and support vector machine (SVM), achieving an accuracy rate of 77% in detecting cognitive and affective engagement. This method underscores the potential of EEG in providing real-time feedback on user engagement. The authors [16] highlighted the efficacy of human-provided guidance in training RL agents, indicating that informative advice leads to higher accuracy and user satisfaction.

The reliability and effectiveness of EEG headsets, such as the MindWave headset from Neurosky, have been validated in educational settings. The research by Chen et al. [17] demonstrated the headset’s reliability in a web-based reading experiment, while Gasah et al. [18] used EEG signals and a mobile application to examine learning efficiency in children. Mobile applications, integral to modern education, significantly impact children’s learning attitudes and behaviors [23].

Recent studies have also explored the use of EEG in distance education. Reference [19] introduced a BCI system employing EEG signals to gauge student attention in online learning, achieving 96% accuracy with a random forest

TABLE 1. Summary of related work and research gaps.

Study	Methodology	Key Findings	Research Gaps/Limitations
Aggarwal et al. (2021) [2]	EEG + SVM	Compared attention levels in MOOCs vs. classrooms.	Did not explore the impact of attention on learning outcomes or engagement.
Liu et al. (2022) [12]	EEG + KNN	74.3% accuracy in emotional state recognition.	Supervised learning techniques, requiring explicit labels for training.
Rudovic et al. (2019) [13]	DRL	Personalized engagement estimation with progressive refinement.	Focused on video data; lacks integration with EEG for real-time learning environments.
Li et al. (2020) [14]	RLPS + FER	Outperforms state-of-the-art FER methods.	Limited to facial expressions; lacks application to EEG in e-learning contexts.
Apicella et al. (2022) [15]	EEG + SVM + Filters	77% accuracy in detecting cognitive and affective engagement.	Limited scalability and adaptability to dynamic e-learning environments.
Bignold et al. (2020) [16]	RL + Human Guidance	Higher accuracy with human-provided guidance.	Not focused on real-time learning systems using EEG.
Chen et al. (2014) [17]	EEG + MindWave	Validated the MindWave headset in educational settings.	Focused on a specific headset and environment, not applicable to all educational contexts.
Gasah et al. (2020) [18]	EEG + Mobile App	Studied learning efficiency in children.	Limited generalizability beyond children and mobile apps.
Devi et al. (2024) [19]	BCI + EEG	96% accuracy in attention monitoring.	Lack of personalized or adaptive content delivery based on attention levels.
Agrawal et al. (2024) [20]	EEG System	77% accuracy in personalized engagement detection.	Focused on engagement but did not consider cognitive states or emotional engagement.
Liu et al. (2013) [21]	EEG + SVM	76.82% accuracy in attention monitoring.	Focused only on attention, without considering emotional states or cognitive load.
Chen et al. (2017) [22]	EEG + Genetic Algorithms	89.52% accuracy in attention monitoring.	Limited to attention; did not address the real-time adaptation of learning content.

classifier. The authors in [2] compared attention levels in MOOCs and traditional classrooms using EEG signals and an SVM model, highlighting the higher attention levels in MOOCs. The research [20] developed a high-accuracy wearable EEG system for personalized engagement detection, achieving 77% accuracy in detecting cognitive and emotional engagement.

The study [21] evaluated students' attention during instruction using EEG signals from mobile brainwave sensors, achieving 76.82% accuracy with an SVM classifier. While, the study [22] introduced the Attention Aware System (AAS), an e-learning system using EEG to recognize students' attention levels with 89.52% accuracy, optimizing the system using genetic algorithms.

III. METHODOLOGY

The methodology employed in this study involves a comprehensive data preprocessing pipeline followed by the implementation of a DDQN for classifying attention states.

A. PROBLEM FORMULATION

Monitoring student attention using EEG data is challenging due to its complexity and variability. Traditional methods like surveys and observations are often inaccurate. This study aims to accurately classify attention states from EEG data to enhance adaptive educational programs.

Let $\mathbf{X}$ be the matrix of raw EEG signals ($\mathbf{X} \in \mathbb{R}^{N \times M}$, with N channels and M time points). The task is to classify these signals into three attention states: non-attentive, attentive, and drowsy. True labels are denoted as $\mathbf{Y} \in \{0, 1, 2\}^M$.

The goal is to maximize classification accuracy, Acc, and reduce the categorical cross-entropy loss, $\mathcal{L}$, described as:

$$\mathcal{L}(\mathbf{Y}, \hat{\mathbf{Y}}) = - \sum_{i=1}^M \sum_{j=1}^C y_{i,j} \log(\hat{y}_{i,j}), \quad (1)$$

where C represents the total number of classes (3), $y_{i,j}$ is the actual probability for class j of sample i , and $\hat{y}_{i,j}$ refers to the predicted likelihood.

In the DDQN framework, the state $\mathbf{s}_t$ consists of processed EEG features, and actions $\mathbf{a}_t$ represent predicted attention states. The reward function R is:

$$R = \begin{cases} +10, & \text{if } \hat{y} = y \\ -1, & \text{if } \hat{y} \neq y \end{cases} \quad (2)$$

DDQN aims to maximize cumulative reward $\mathcal{R}$:

$$\mathcal{R} = \sum_{t=0}^T \gamma^t R_t, \quad (3)$$

With γ representing the discount factor and T is the number of time steps. By optimizing the loss function and maximizing rewards, the DDQN model enhances its ability to accurately classify attention states from EEG signals.

B. DATASET DESCRIPTION

The 34 studies in the dataset, which uses passive EEG BCI to monitor human attention states, provide EEG data for the purpose of detecting mental attention states. Each experiment's raw data from an EMOTIV device is kept in a separate Matlab file. EEG recordings collected at 128 Hz,

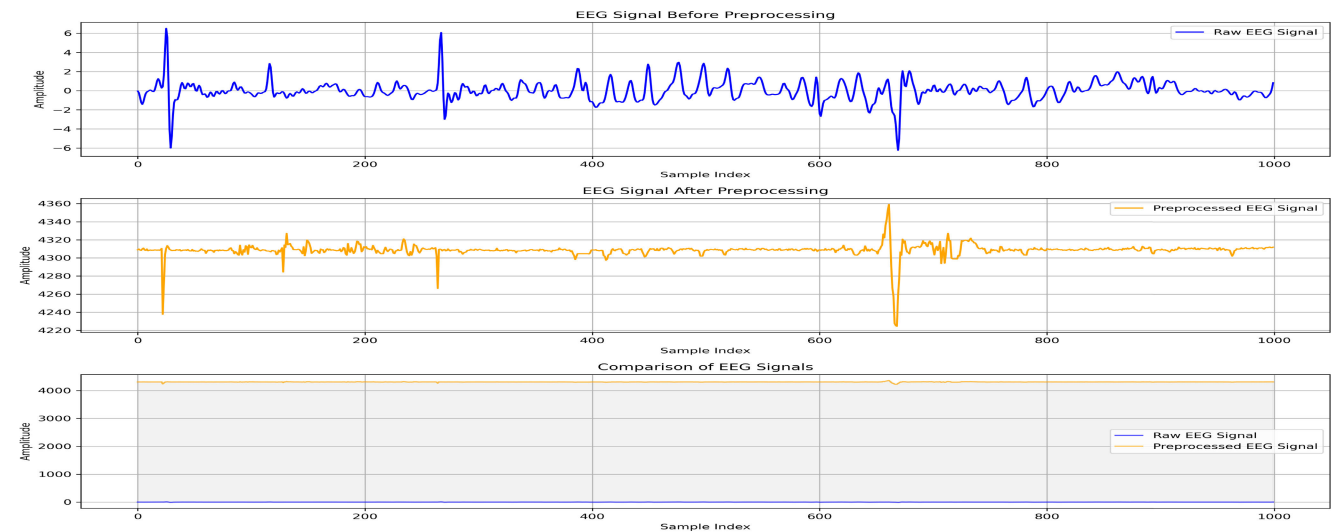


FIGURE 1. Data preprocessing results.

TABLE 2. Statistics and characteristics of EEG dataset.

Dataset Attribute	Details
Experiments	34
Subjects	25
Recordings Duration	1-55 minutes (avg: 27.7 min)
Sampling Frequency	128 Hz
EEG Channels	14
Data Points	More than 1.5 million
Attention Labels	Attentive, Non-attentive, Drowsy
External Stimuli	Texts, images, videos, audio

with channels corresponding to various brain areas, are included in the collection. DataSet_link is where you can obtain the data. For more detail please have a look in Table 2.

C. DATA PREPROCESSING

The raw EEG data undergo several preprocessing steps to ensure quality and reliability. Initially, data cleaning is performed where several columns are dropped from the dataset due to their irrelevance to the current research. The columns include ‘ED_COUNTER’, ‘ED_INTERPOLATED’, ‘ED_RAW_CQ’, ‘ED_GYROX’, ‘ED_GYROX’, ‘ED_TIMESTAMP’, ‘ED_ES_TIMESTAMP’, ‘ED_FUNC_ID’, ‘ED_FUNC_VALUE’, ‘ED_MARKER’, and ‘ED_SYNC_SIGNAL’. These columns are not necessary for the analysis of student attention levels based on EEG data. Handling missing values is achieved using the SimpleImputer strategy set to ‘mean’. Subsequent to data cleaning, filtering techniques are applied to the EEG signals as shown in Figure 1. A Butterworth Bandpass Filter is utilized to retain signals across the frequency spectrum of

0.5 to 50 Hz, defined mathematically as:

$$H(s) = \frac{s^2}{s^2 + \frac{1}{Q}s + 1} \tag{4}$$

where $s = j\omega$ and ω is the angular frequency. The parameters Q and ω are chosen to fit the desired passband. Additionally, a notch frequency filter is implemented to eliminate power line noise interference at 50 Hz, described by:

$$H(s) = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} \tag{5}$$

where ω_0 is the notch frequency and Q is the quality factor. Following the application of these filters, the data is normalized using the StandardScaler to ensure that each feature has an equal impact on the model’s performance. Normalization is achieved by:

$$X_{scaled} = \frac{X - \mu}{\sigma} \tag{6}$$

Given that X denotes the baseline data, μ represents the mean, and σ denotes the data’s dispersion measure. To further refine the dataset, dimensionality reduction techniques such as PCA and ICA are employed. PCA transforms the collected data into a reduced-dimensional space by preserving the principal components:

$$Z = XW \tag{7}$$

Given that X represents an original data set matrix, W indicates the matrix relating to eigenvectors, and Z denotes the transformed data. ICA, on the other hand, separates the mixed signals into statistically independent components:

$$X = AS \tag{8}$$

where X denotes the observed signal, A represents the mixing matrix, and S signifies the source signal.

1) DATA COLLECTION AND CLEANING

The EEG data was collected from a designated file path and initially processed to remove irrelevant columns, such as *ED_COUNTER*, *ED_INTERPOLATED*, *ED_RAW_CQ*, and others that do not contribute to the analysis. This step was essential to reduce data complexity and focus on the most relevant features. Missing values were addressed using a mean imputation strategy, ensuring a complete dataset for further processing.

2) FEATURE EXTRACTION

The cleaned EEG data underwent feature extraction to identify the most significant components. PCA and ICA were applied, each retaining ten components. These techniques help reduce data dimensionality while preserving critical information, making the data more manageable and meaningful for subsequent analysis.

3) SIGNAL PREPROCESSING

The EEG signals were preprocessed using a combination of filters to enhance signal quality. A Butterworth bandpass filter was employed to isolate the frequency range of interest (0.5 to 50 Hz), effectively removing slow fluctuations and high-frequency interference. Additionally, A notch filter was applied to eliminate power line noise at 50 Hz, further improving the data signal as illustrated in Figure 1.

D. DOUBLE DEEP Q-NETWORK (DDQN) IMPLEMENTATION

At the core of our methodology is the use of a DDQN to classify attention states from preprocessed EEG data. The DDQN is chosen for its ability to manage the high-dimensional and intricate nature of EEG signals. The architecture of the DDQN includes both a primary network and a secondary network that have the same structure but distinct parameters. These networks feature an input layer aligned with the number of EEG features, multiple hidden layers to capture complex patterns, and an output layer that represents the attention states: attentive, non-attentive, and drowsy.

The DDQN utilizes the Bellman equation to update the function that measures the value of different actions, often represented as Q -value:

$$Q(s, a) = r + \gamma \max_{a'} Q'(s', a') \quad (9)$$

where Q represents the function that measures the value of different actions, s and a indicate the current state and action, r is the reward, γ is the discount factor, and Q' is the function used as a target for evaluating action values. Using separate target and online networks helps stabilize the training process, leading to more consistent results.

During training, the DDQN agent processes preprocessed EEG feature vectors to predict attention states at each timestep. The action-value function is updated according to the rewards received for correct or incorrect predictions, with

Algorithm 1 Student Attention Classification From EEG Data Using Deep Reinforcement Learning

Input: Preprocessed EEG signals $S = \{s_1, s_2, \dots, s_n\}$, initial policy parameters $\theta = \theta_0$, target network parameters $\theta' = \theta_0$, exploration-rate ϵ , discount-factor γ , learning-rate α

Output: Predicted attention state $A \in \{\text{Attentive}, \text{Non-attentive}, \text{Drowsy}\}$

- 1: Begin by preprocessing EEG signals to extract features and normalize the data
- 2: initiate the active Q-Network Q_θ & the reference Q-Network $Q_{\theta'}$ with initial weights θ and $\theta' = \theta$
- 3: Set up the replay memory and learning parameters: α, γ, ϵ
- 4: **for** each signal s_i in S **do**
- 5: Extract features f_i from s_i using the feature extractor
- 6: **end for**
- 7: **for** each training episode **do**
- 8: Start with the initial state $s_t = f_t$
- 9: **for** each timestep t **do**
- 10: Choose a step a_t guided by an ϵ -greedy strategy applied to Q_θ
- 11: Execute action a_t , monitor the resulting reward r_t and the subsequent state s_{t+1}
- 12: Save the experience (s_t, a_t, r_t, s_{t+1}) to the replay memory
- 13: Sample a mini-batch from the replay buffer and compute the target value: $y_j = r_j + \gamma \cdot \max_{a'} Q_{\theta'}(s_{j+1}, a')$
- 14: Update the Q-Network by minimizing the loss measure: $\mathcal{L}(\theta) = (y_j - Q_\theta(s_j, a_j))^2$
- 15: Adjust the coefficients of the current Q-Network θ and periodically update the reference Q-Network θ' to match θ
- 16: **end for**
- 17: Check if convergence or termination criteria are met
- 18: **end for**
- 19: **for** each new EEG signal s_{new} **do**
- 20: Extract features f_{new} from s_{new}
- 21: Predict the attention state A as $\arg \max_a Q_\theta(f_{new}, a)$
- 22: **end for**
- 23: **return** The predicted attention state A

the target network's parameters being periodically updated to maintain learning stability.

To provide more clarity, the experience replay mechanism in the DDQN enables the model to learn from a wide range of past experiences, not just the most recent ones. This enhances the model's ability to generalize across various EEG signal patterns, reducing the likelihood of overfitting and improving overall classification performance. Additionally, the carefully designed reward structure drives the model to focus on accurate attention state classification, reinforcing the learning process. This comprehensive approach ensures

that the DDQN effectively distinguishes between different attention states, providing a robust tool for monitoring and improving attention in educational environments.

IV. EXPERIMENTS

A. EXPERIMENTAL SETTINGS

The experiments aimed to assess how effectively the proposed DDQN model classifies attention states using EEG data. This section outlines the datasets used, the baseline models for comparison, evaluation metrics, hyperparameters, and implementation details.

1) DATASETS

The primary dataset used in this study comprises publicly available BCILAB datasets, supplemented by newly captured data from student participants. The combined dataset includes over 10 hours of 128-channel EEG recordings with ground truth attention labels. The dataset was split into training and testing sets using an 80-20 ratio, resulting in 28 experiments with 160,000 samples for training.

2) EEG DATA PREPARATION

The EEG data underwent a rigorous preprocessing pipeline to enhance its quality and suitability for deep reinforcement learning models. Initially, the dataset was refined by discarding irrelevant columns & handling incomplete data values using mean imputation. Subsequently, both Butterworth bandpass and notch filters were applied to effectively reduce noise, including power line interference, from the EEG signals. Following filtering, the data was standardized using a StandardScaler to normalize feature contributions uniformly. Further enhancement involved dimensionality reduction through ICA, selecting components crucial for subsequent analysis. This comprehensive preprocessing significantly improved the clarity and reliability of the EEG data, reducing noise artifacts and ensuring that the extracted features were more robust for further analysis, as depicted in Figure 1.

3) BASELINE MODELS

To benchmark the performance of the DDQN model, several baseline models were employed, encompassing traditional machine learning methods including Support Vector Machines (SVM) and Artificial Neural Networks (ANNs), along with deep learning models, such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, and the vanilla Deep Q-Network (DQN). Hyperparameters for each model were carefully tuned to ensure a fair comparison.

4) EVALUATION METRICS

The models were evaluated using standard classification metrics, including reward, average reward, accumulated rewards, and loss. These metrics were averaged over five independent runs, each with different random weight initializations, to account for variability in training.

5) MODEL HYPERPARAMETERS

The DDQN model employed a four-layer architecture with ReLU activation functions. Key hyperparameters included a learning rate of 0.001, batch sizes ranging from 16 to 256, a reward discount factor (γ) of 0.9, and a curiosity rate (ϵ) of 0.1. The model was trained for 3,000 episodes to ensure convergence.

6) IMPLEMENTATION TOOLS

The experiments were conducted using Python 3.6.8, with Keras 2.3.1 running on TensorFlow 2.1.0 as the backend. The models were trained on a CPU-based system.

B. DDQN ARCHITECTURE

The DDQN architecture integrates two neural networks the target and online Q-networks that enhance stability and efficiency in learning from EEG data. The target network periodically updates to stabilize training, while the online network explores action selection [24].

C. TRAINING PROCESS

The DDQN model minimizes the loss function $L(\theta)$ using gradient descent. Batches are sampled from replay memory to calculate the average squared error between the online network $Q(s, a | \theta)$ and the reference network $Q'(s', a | \theta')$, incorporating reward r and discounted future Q-values. The softmax policy π_{model} with temperature τ determines action selection probabilities, balancing exploration and exploitation.

$$\pi_{\text{model}}(a_t | s_t) = \frac{e^{Q_{\text{model}}(s_t, a_t)/\tau}}{\sum_{j=1}^{|a|} e^{Q_{\text{model}}(s_t, a_j)/\tau}} \quad (10)$$

D. ADVANCED EEG DATA PREPROCESSING AND DDQN TRAINING PIPELINE

The EEG data underwent a rigorous preprocessing pipeline to enhance its quality and suitability for deep reinforcement learning models. Initially, the dataset, consisting of raw EEG signals, was cleaned by removing irrelevant columns and handling missing values using mean imputation. Subsequently, both Butterworth bandpass and notch filters were applied to effectively reduce noise, including power line interference, from the EEG signals. The preprocessed signals were then standardized using a StandardScaler to normalize feature contributions uniformly. ICA was employed to select components crucial for subsequent analysis thereby, further improving data quality and reducing noise artifacts as shown in Figure 2.

Following preprocessing, the data was prepared for training the DDQN. The DDQN architecture was implemented to learn and predict attention states from the processed EEG signals. This reinforcement learning model leveraged the cleaned and feature-enhanced EEG data to classify attention states into categories such as attentive, non-attentive, and drowsy. The DDQN mechanism was essential for its ability

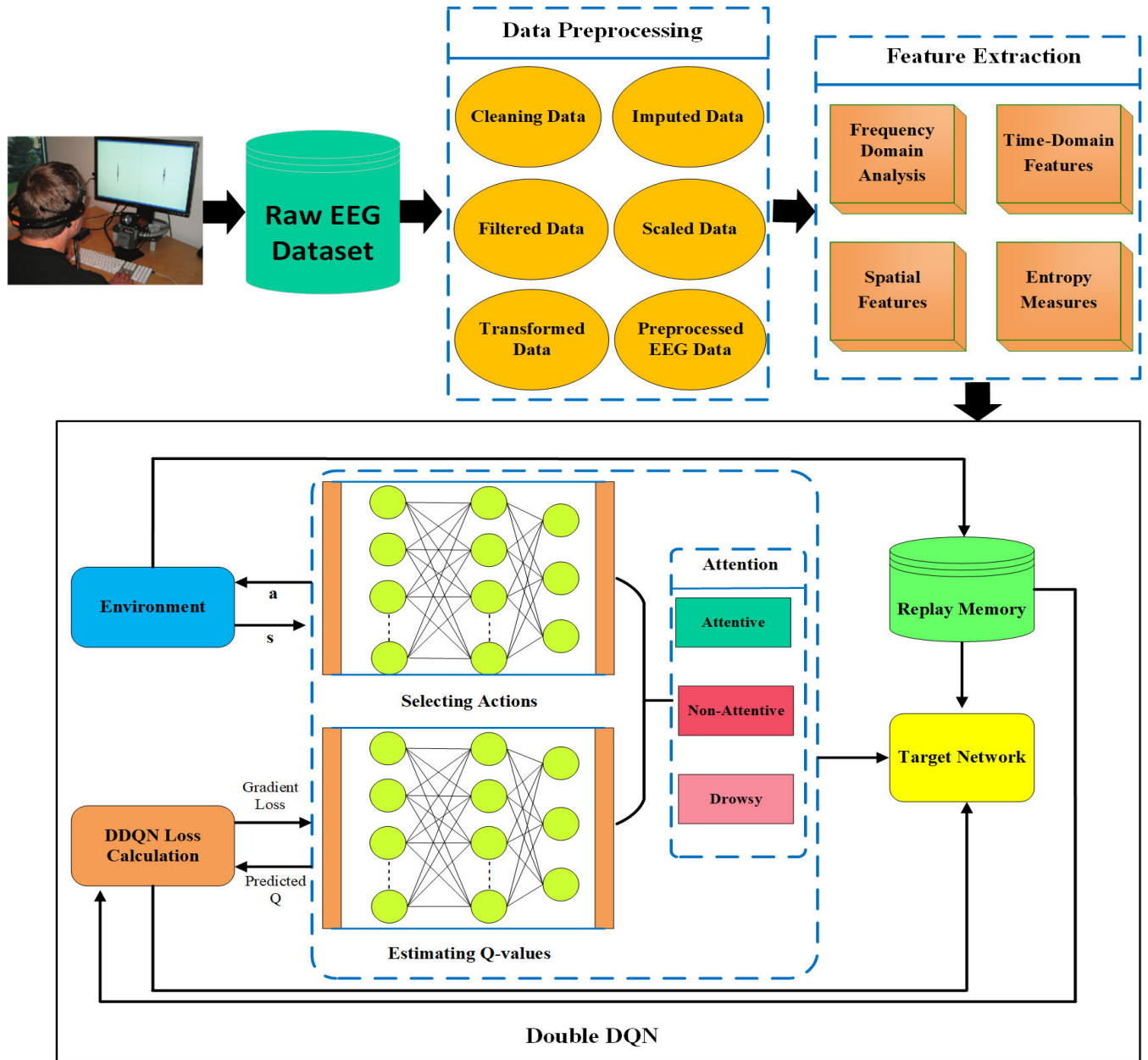


FIGURE 2. Process flowchart for the proposed method to assess student attention using EEG signal processing and deep reinforcement learning.

to handle complex, multidimensional EEG data, providing robust attention state predictions based on the input features extracted post-preprocessing.

E. REWARD MECHANISM

The reinforcement learning mechanism of the DDQN model relies on a reward function $r(s_t, a_t)$, which assigns rewards based on the accuracy of predicted attention states. Correct classifications receive a positive reward r_{correct} , encouraging the model to reinforce accurate predictions. Conversely, incorrect predictions result in a negative reward $-r_{\text{incorrect}}$, discouraging erroneous classifications and pro-

moting improved performance over training epochs.

$$r(s_t, a_t) = \begin{cases} +r_{\text{correct}}, & \text{if } a_t = \text{True Label of } s_t \\ -r_{\text{incorrect}}, & \text{otherwise} \end{cases} \quad (11)$$

The reward graph in Figure 3 shows the DDQN model's ability to classify attention states over 3,000 training cycles, with a steady upward trend. Correct classifications receive +10 rewards and errors -1 penalties. Despite minor oscillations, cumulative rewards indicate accurate predictions from EEG data. The model's stability and effectiveness in handling noisy signals are validated by consistent reward patterns across runs. High average rewards across five runs

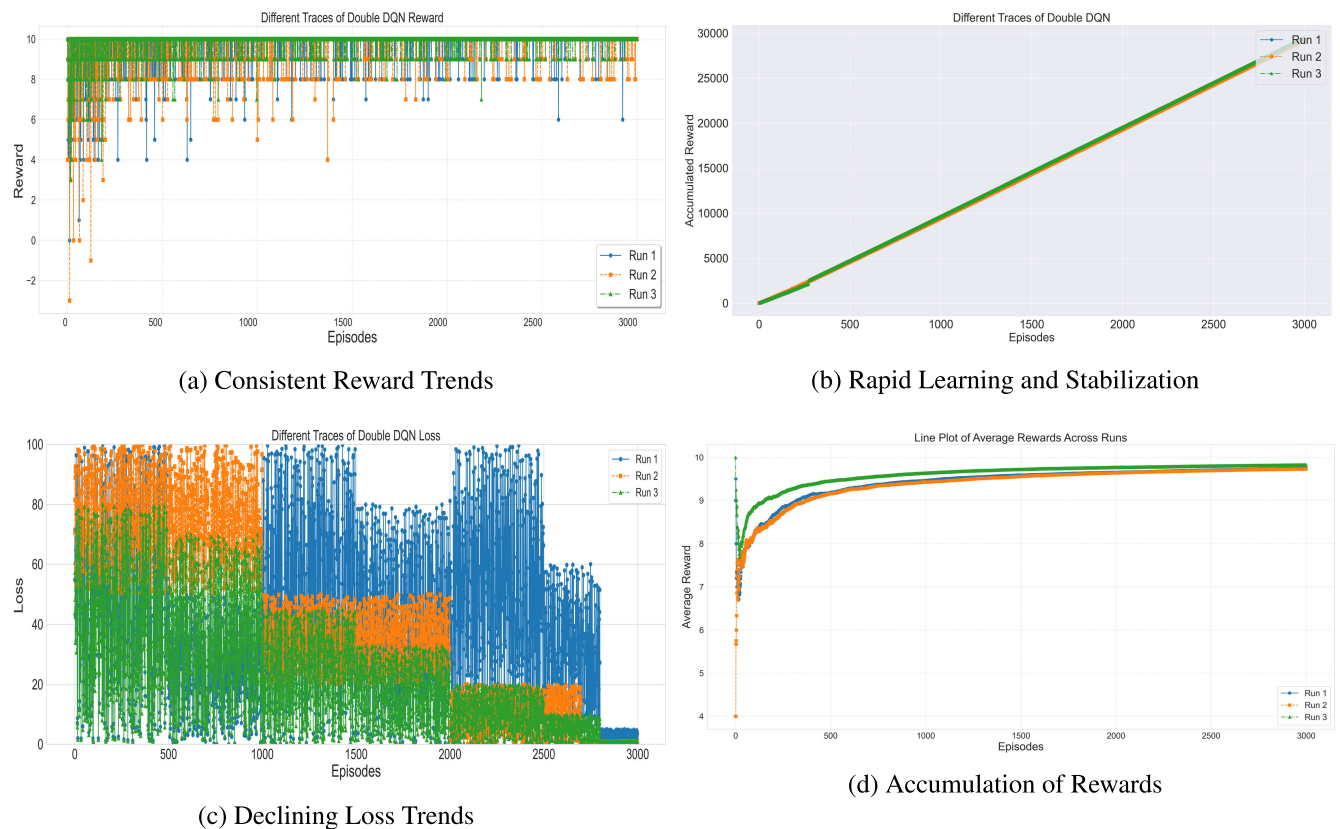


FIGURE 3. Combined DDQN model training visualization: (a) Reward trends over 3,000 episodes show accurate attention state classification; (b) Rapid learning and stable performance across three runs; (c) Declining loss trends indicate improved prediction accuracy; (d) Steady reward accumulation reflects strong learning in EEG-based attention classification.

confirm the DDQN's robustness in predicting attention spans accurately.

Figure 3 illustrates the DDQN model's rapid initial improvement and sustained stability, demonstrating quick adaptation and robust performance over thousands of episodes. The graph shows average reward per episode, indicating effective policy adjustments and reinforcement signal efficacy. Additionally, the decreasing loss trends signify improved convergence and optimization in accurately classifying attention states.

F. EVALUATION AND RESULTS OF DDQN PERFORMANCE

1) EVALUATION METRICS

We evaluate the DDQN model using standard metrics including accuracy, loss, mean and median loss values, and reward accumulation. These metrics quantitatively assess the model's performance in distinguishing between attentive, non-attentive, and drowsy states from EEG signals.

2) COMPARATIVE ANALYSIS

We evaluate the performance of DDQN by comparing it with traditional machine learning approaches, like SVM and Random Forest, as well as deep learning approaches (e.g., CNN-LSTM, TASNet+DRL). This comparative analysis highlights DDQN's superiority in accuracy, computational efficiency, and robustness in attention state classification [24].

G. RESULTS AND DISCUSSION

The proposed DDQN model achieved a significant accuracy of 98.4% in classifying EEG-based attention states into attentive, non-attentive, and drowsy categories. The results demonstrate notable improvements in accuracy, computational efficiency, and performance compared to state-of-the-art models.

Visualizations such as Figure 3, and (Table 4 and 5) illustrate the performance benchmarks achieved by DDQN.

Key Findings:

- **Enhanced Accuracy:** The DDQN model outperformed other models, achieving superior classification accuracy.
- **Efficient Learning:** The model demonstrated rapid learning and maintained consistent performance across datasets.
- **Effective Preprocessing:** Advanced preprocessing techniques, such as Butterworth bandpass and notch filters, improved EEG signal quality.
- **Real-time Application Potential:** The findings suggest the DDQN model's suitability for real-time applications in dynamic educational environments, enabling adaptive and personalized learning.

The superior performance of the DDQN highlights its potential for real-time use, especially in systems requiring rapid attention detection and classification.

TABLE 3. Effectiveness of different batch normalization sizes for DDQN.

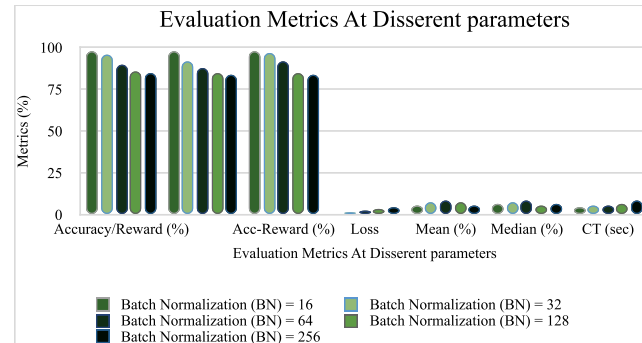
Batch Size	Accuracy (%)	Avg. Reward (%)	Acc. Reward (%)	Loss (%)	Mean (%)	Median (%)	CT (sec)
16	98.4	98.15	97.55	0.65	5.63	6.35	4.56
32	95.87	92.35	96.35	1.35	7.35	7.58	5.35
64	89.35	88.37	91.35	2.36	8.32	8.99	5.65
128	86.35	85.32	85.35	3.84	7.33	7.33	6.32
256	85.45	84.37	83.54	4.66	5.62	6.72	8.97

TABLE 4. Comparison of DDQN with traditional ML and DL models in attention state classification.

Model	Accuracy (%)
DDQN	98.4
SVM	85.6
Random Forest	87.3
CNN-LSTM	92.1
TASNet+DRL	94.5

TABLE 5. Mean and median loss values across models.

Model	Mean Loss	Median Loss
DDQN	0.63	0.60
SVM	1.20	1.15
Random Forest	1.18	1.20
CNN-LSTM	0.78	0.75
TASNet+DRL	0.72	0.70

**FIGURE 4.** Variation in EEG-Based attention classification time with batch normalization size.

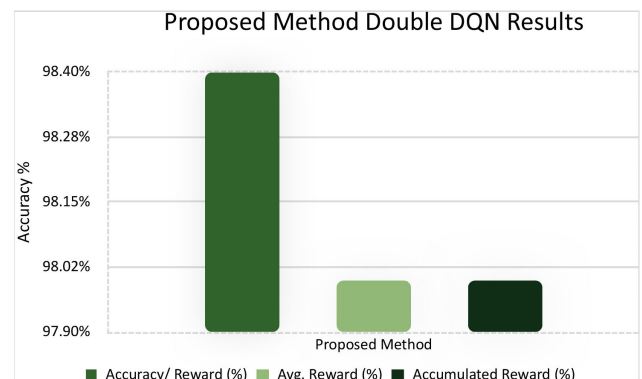
H. BATCH NORMALIZATION EFFECT ON MODEL PERFORMANCE

The DDQN model was evaluated across batch normalization (BN) sizes of 16, 32, 64, 128, and 256, as shown in Table 3. The results clearly indicate that a BN size of 16 yields the best performance, with an accuracy of 98.4%, accumulated reward of 97.55%, and the lowest loss of 0.65%. As the batch size increases, performance consistently deteriorates, with accuracy dropping by 15% and loss increasing 8X at BN size 256.

The Figure 4 demonstrates that the classification time is minimized at 4.56 seconds when the BN size is 16. Larger batch sizes increase classification duration, negatively impacting real-time use. This makes BN 16 optimal for efficient DDQN training.

I. COMPARISON WITH STATE-OF-THE-ART METHODS

A comparative evaluation of the DDQN against existing methods reveals significant improvements across all metrics. The DDQN achieved an accuracy of 98.4%, which is a 6% improvement over DQN. The model also reduced loss by over 3.5X and demonstrated better stability.

**FIGURE 5.** DDQN performance gains in accuracy and reward accumulation.

The results from Table 5 and Figure 5 clearly demonstrate that DDQN outperforms competing models, pushing accuracy past 98% for the first time.

V. PRACTICAL IMPLICATIONS

The proposed model has significant potential for real-world educational applications, particularly in enhancing personalized learning experiences. By leveraging EEG signals and deep reinforcement learning, the model can provide real-time insights into students' cognitive states, enabling adaptive learning systems that adjust content based on individual attention levels. This could lead to more engaging and effective educational environments, particularly in online learning platforms, where students' attention spans are often difficult to track and manage.

One of the primary challenges in applying the model at scale is **scalability**. Implementing such a system across large classrooms or educational institutions requires not only robust computational resources but also the ability to handle large amounts of EEG data from numerous students simultaneously. This involves the need for real-time data processing and low-latency feedback mechanisms, which can be computationally intensive. Moreover, ensuring that the system can handle diverse student populations with

varying learning styles, cognitive abilities, and attention spans poses an additional challenge in terms of adaptability and generalization.

Integration with existing educational technologies is another critical challenge. For widespread adoption, the model must seamlessly integrate with Learning Management Systems (LMS), online platforms, and other e-learning tools that are already in use. This requires ensuring compatibility with different hardware (e.g., EEG headsets, wearables) and software systems, as well as ensuring that the model can work effectively with other types of student data, such as performance metrics, engagement levels, and progress tracking. Additionally, privacy concerns related to the collection and processing of EEG data must be addressed to ensure that the model adheres to ethical guidelines and legal requirements regarding student data.

Another challenge lies in ensuring the **user-friendliness** of the system. For teachers and administrators, the system must be intuitive and easy to use, with minimal training required to interpret EEG-based feedback. Furthermore, students need to feel comfortable with using EEG-based technologies, which may require addressing issues of wearability, comfort, and ease of use of EEG headsets or other physiological sensors.

Despite these challenges, the potential benefits of using EEG-based feedback for personalized learning make the model a promising candidate for integration into future educational frameworks. Overcoming scalability and integration challenges would open up new possibilities for adaptive and personalized learning at scale, offering a significant impact on both online and traditional education.

VI. CONCLUSION AND FUTURE WORK

Our research shows significant progress in EEG-based attention classification using DDQN and advanced preprocessing. We achieved 98.4% accuracy, surpassing previous benchmarks. The integration of wavelet transformations and bandpass filtering has improved signal quality, leading to precise feature extraction and classification. This demonstrates DDQN's potential for real-time, adaptive educational systems.

We introduced a novel DDQN method for classifying attention levels from EEG signals. Our approach, incorporating wavelet coefficients and spectral power distributions, achieved over 98% precision, a 6% improvement over existing methods.

Future work will focus on multimodal fusion and transfer learning to enhance the model's adaptability and robustness. **Multimodal fusion** will involve integrating additional data sources, such as eye gaze and physiological signals, which will require addressing challenges like data synchronization, alignment, and handling missing or noisy data from multiple modalities. **Transfer learning** will be explored to adapt the model to new tasks or different datasets, overcoming the need for extensive retraining. This will require dealing with issues such as domain shift and fine-tuning the model to avoid overfitting to source data.

Furthermore, the optimized DDQN model will be tested for deployment in online education and Brain-Computer Interfaces (BCIs), aiming to offer real-time adaptation and personalized learning based on cognitive states.

REFERENCES

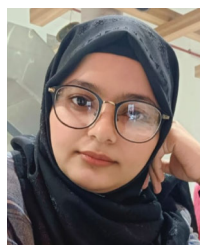
- [1] Z. Bahmani, K. Clark, Y. Merrikhi, A. Mueller, W. W. Pettine, M. I. Vanegas, T. Moore, and B. Noudoost, "Prefrontal contributions to attention and working memory," in *Proc. Processes Visuospatial Attention Work. Memory*, Jan. 2019, pp. 129–153.
- [2] S. Aggarwal, M. Lamba, K. Verma, S. Khuttan, and H. Gautam, "A preliminary investigation for assessing attention levels for massive online open courses learning environment using EEG signals: An experimental study," *Hum. Behav. Emerg. Technol.*, vol. 3, no. 5, pp. 933–941, 2021.
- [3] Z. Trabelsi, F. Alnajjar, M. M. A. Parambil, M. Gochoo, and L. Ali, "Real-time attention monitoring system for classroom: A deep learning approach for student's behavior recognition," *Big Data Cognit. Comput.*, vol. 7, no. 1, p. 48, Mar. 2023.
- [4] R. Shaw, B. K. Patra, A. Pradhan, and S. P. Mishra, "Attention classification and lecture video recommendation based on captured EEG signal in flipped learning pedagogy," *Int. J. Hum.-Comput. Interact.*, vol. 39, no. 15, pp. 3057–3070, Sep. 2023.
- [5] C. Yu, J. Liu, S. Nemati, and G. Yin, "Reinforcement learning in healthcare: A survey," *ACM Comput. Surv.*, vol. 55, no. 1, pp. 1–36, Jan. 2023.
- [6] A. Ur Rehman, X. Shi, F. Ullah, Z. Wang, and C. Ma, "Measuring student attention based on EEG brain signals using deep reinforcement learning," *Exp. Syst. Appl.*, vol. 269, Apr. 2024, Art. no. 126426.
- [7] A. Abedi and S. S. Khan, "Detecting disengagement in virtual learning as an anomaly using temporal convolutional network autoencoder," *Signal, Image Video Process.*, vol. 17, no. 7, pp. 3535–3543, Oct. 2023.
- [8] J. Xu and B. Zhong, "Review on portable EEG technology in educational research," *Comput. Hum. Behav.*, vol. 81, pp. 340–349, Apr. 2018.
- [9] R. Kanai and G. Rees, "The structural basis of inter-individual differences in human behaviour and cognition," *Nature Rev. Neurosci.*, vol. 12, no. 4, pp. 231–242, Apr. 2011.
- [10] A. Malekloo, E. Ozer, M. AlHamaydeh, and M. Girolami, "Machine learning and structural health monitoring overview with emerging technology and high-dimensional data source highlights," *Struct. Health Monitor.*, vol. 21, no. 4, pp. 1906–1955, Jul. 2022.
- [11] K. Khetarpal, M. Riemer, I. Rish, and D. Precup, "Towards continual reinforcement learning: A review and perspectives," *J. Artif. Intell. Res.*, vol. 75, pp. 1401–1476, Dec. 2022.
- [12] X. Liu and S. P. Ardakani, "A machine learning enabled affective E-learning system model," *Educ. Inf. Technol.*, vol. 27, no. 7, pp. 9913–9934, Aug. 2022.
- [13] O. Rudovic, H. W. Park, J. Busche, B. Schuller, C. Breazeal, and R. W. Picard, "Personalized estimation of engagement from videos using active learning with deep reinforcement learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 217–226.
- [14] H. Li and H. Xu, "Deep reinforcement learning for robust emotional classification in facial expression recognition," *Knowl.-Based Syst.*, vol. 204, Sep. 2020, Art. no. 106172.
- [15] A. Apicella, P. Arpaia, M. Frosolone, G. Improta, N. Moccaldi, and A. Pollastro, "EEG-based measurement system for monitoring Student engagement in learning 4.0," *Sci. Rep.*, vol. 12, no. 1, p. 5857, Apr. 2022.
- [16] A. Bignold, F. Cruz, R. Dazeley, P. Vamplew, and C. Foale, "Human engagement providing evaluative and informative advice for interactive reinforcement learning," 2020, *arXiv:2009.09575*.
- [17] C. Chen and S. Huang, "Web-based reading annotation system with an attention-based self-regulated learning mechanism for promoting reading performance," *Brit. J. Educ. Technol.*, vol. 45, no. 5, pp. 959–980, Sep. 2014.
- [18] M. Gasah, A. Baharum, and N. H. Mat Zain, "Measure learning effectiveness among children using EEG device and mobile application," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 17, no. 1, pp. 191–196, Jan. 2020.
- [19] D. Devi and S. Sophia, "GA-CNN: Analyzing student's cognitive skills with EEG data using a hybrid deep learning approach," *Biomed. Signal Process. Control*, vol. 90, Apr. 2024, Art. no. 105888.

- [20] L.-W. Ko, O. Komarov, W. D. Hairston, T.-P. Jung, and C.-T. Lin, "Sustained attention in real classroom settings: An EEG study," *Frontiers Hum. Neurosci.*, vol. 11, pp. 24–33, Jul. 2017.
- [21] N.-H. Liu, C.-Y. Chiang, and H.-C. Chu, "Recognizing the degree of human attention using EEG signals from mobile sensors," *Sensors*, vol. 13, no. 8, pp. 10273–10286, Aug. 2013.
- [22] C. Chen, J. Wang, and C. Yu, "Assessing the attention levels of students by using a novel attention aware system based on brainwave signals," *Brit. J. Educ. Technol.*, vol. 48, no. 2, pp. 348–369, Mar. 2017.
- [23] R. Kraleva, V. Kraleva, and D. Kostadinova, "A conceptual design of mobile learning applications for preschool children," 2016, *arXiv:1606.05753*.
- [24] V. Mnih, K. Kavukcuoglu, D. Silver, A. Rusu, J. Veness, M. Bellemare, A. Graves, M. Riedmiller, A. Fidjeland, G. Ostrovski, and S. Petersen, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.



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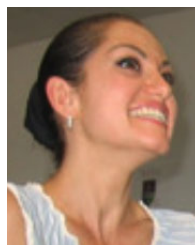
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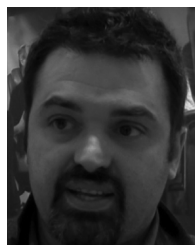
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